Neural CCTV: An Autonomous Edge-Native Video Analytics Architecture Integrating Hardware Motion Gating, Multimodal Foundation Models, and Vector Search

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Abstract—Continuous high-resolution video surveillance imposes severe computational and thermal burdens on edge appliances, while offloading video feeds to public cloud services introduces recurring subscription expenses, network bandwidth saturation, and critical privacy vulnerabilities. Monitored CCTV feeds remain quiescent for upwards of 85%–95% of their operational lifespan, yet conventional analytics frameworks continuously burn GPU resources evaluating redundant frames. To address these challenges, this paper presents Neural CCTV, a modular, edge-native, zero-cloud forensic video analytics architecture. The system orchestrates five decoupled tiers: (i) an ultra-fast hardware-accelerated motion event clipper utilizing NVIDIA NVDEC and vectorized difference calculations to scan 2K footage at $> 1,000$ FPS and losslessly extract active events; (ii) a quantized Vision-Language Model (VLM) engine based on Qwen3-VL-4B-Instruct running in 4-bit NormalFloat4 (NF4) with Scaled Dot-Product Attention (SDPA) within a 3.4 GB VRAM envelope, generating structured forensic activity logs; (iii) a Chronological Visual Retrieval-Augmented Generation (Visual RAG) protocol that maintains temporal continuity, cross-referencing multi-clip entities to reliably detect status and cargo transitions (e.g., empty delivery vehicles departing loaded); (iv) a multimodal semantic search engine pairing Google VideoPrism 768-dimensional video-to-vector embeddings with an embedded ChromaDB vector database using Hierarchical Navigable Small World (HNSW) graphs for sub-millisecond natural language video retrieval; and (v) an offline biometric and spatial verification tier combining MTCNN, Inception-ResNet-V1, and polygonal geofencing. Comprehensive physical evaluations on edge hardware demonstrate a 91.3% reduction in GPU idle compute, 68.2% lower system power consumption, 100% recall on transient events, and instantaneous open-vocabulary forensic query retrieval, delivering total data sovereignty under zero cloud dependency.

Index Terms—Edge Computing, Video Surveillance, Vision-Language Models, VideoPrism, Qwen3-VL, Vector Databases, ChromaDB, Temporal Motion Gating, Visual RAG.

I. Introduction

The ubiquity of high-definition Internet Protocol (IP) closed-circuit television (CCTV) cameras has revolutionized physical perimeter defense, industrial inspection, and residential security [1]. Globally, hundreds of millions of surveillance cameras stream visual telemetry uninterruptedly. However, the volume of raw video data produced daily vastly outstrips human supervisory capacity [2]. Consequently, video surveillance remains fundamentally forensic and reactive: human operators manually scrutinize hours of unindexed recordings only after an anomaly or security breach has taken place.

Modern commercial solutions attempt to automate monitoring by streaming video streams to third-party public clouds for centralized deep learning inference [3]. This paradigm suffers from three critical bottlenecks:

- Privacy and Regulatory Vulnerabilities: Transmitting private domestic or enterprise video telemetry over external networks exposes intimate spaces to data breaches, interception, and regulatory infractions under the EU General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [4].
- High Uplink Bandwidth and Recurring Subscriptions: Streaming multi-camera 2K or 4K feeds demands continuous multi-megabit uplinks (2.5–6.0 Mbps per channel) and imposes perpetual cloud storage and computing fees.
- Alert Fatigue and Cryptic Metadata: Conventional computer vision systems output cryptic bounding boxes, classes, and arbitrary confidence scores, overwhelming operators with false alarms while failing to describe the high-level semantic narrative of the scene.

Conversely, hosting continuous deep neural network inference directly on edge hardware (e.g., standard consumer GPUs or embedded accelerators) encounters thermal throttling, excessive power draw, and GPU memory saturation [5]. Crucially, residential and secure perimeter environments exhibit temporal sparsity: scenes remain completely static for 85% to 95% of their cycle. Evaluating quiescent video frames with heavy deep learning models

represents an enormous waste of electrical energy and compute.

To resolve these compounding challenges, this work introduces Neural CCTV, an autonomous, energy-efficient, edge-native surveillance and forensic video intelligence platform. Running 100% locally without proprietary cloud lock-in, Neural CCTV decouples high-speed temporal motion event extraction from heavy semantic reasoning, and empowers operators with open-vocabulary natural language video retrieval.

A. Key Technical Contributions

The primary contributions of this project are:

1) Hardware-Accelerated Motion Event Clipper ($> 1,000$ FPS): A high-throughput scanning subsystem employing NVIDIA NVDEC on-chip hardware decoding (hevc_cuvid, h264_cuvid) with on-VPU downsampling to 320×180 resolution, vectorized luminance differencing, and lossless FFmpeg stream extraction (-c copy) with dynamic temporal padding.
2) Edge Vision-Language Model Forensic Synthesizer: A local deployment of Qwen3-VL-4B-Instruct quantized to 4-bit NormalFloat4 (NF4) with Scaled Dot-Product Attention (SDPA), achieving deep scene understanding and structured forensic reporting (trajectory, entities, clothing, vehicle cargo) within a 3.4 GB VRAM budget.
3) Chronological Visual RAG for Incident Continuity: A temporal context-injection mechanism that cross-references sequential incident clips within the camera timeline, allowing the VLM to identify entity status changes (e.g., delivery vehicles arriving empty and departing loaded with parcels).
4) Multimodal Vector Database for Natural Language Video Search: Integration of Google VideoPrism 768-dimensional multimodal video-text embeddings with an embedded ChromaDB HNSW vector index, enabling instantaneous (< 1 ms) semantic similarity search over historical video clips using arbitrary natural language queries.
5) Zero-Cloud Biometric and Full-Stack Dashboard: An offline facial verification engine (MTCNN + Inception-ResNet-V1) with active unknown face triage, accompanied by a reactive FastAPI WebSocket backend and React 18 / Vite interactive security HUD.

II. Related Work

A. Efficient Video Processing and Motion Filtering

The computational bottleneck of running continuous deep neural inference on raw surveillance video has motivated two-stage filtering architectures. NoScope [5] and Focus [6] employ specialized small convolutional networks to detect target presence before querying larger models. However, training and updating auxiliary classifiers on edge nodes introduces significant maintenance overhead and model drift. Classical background subtraction algorithms, such as Gaussian Mixture Models (MOG2) [7] and ViBe [8], offer pixel-level motion segmentation but incur substantial CPU memory traffic and suffer from sensitivity to ambient lighting variations. In contrast, our hardware-accelerated clipper executes downsampled frame differencing directly on hardware video processing units (VPUs) and vectorized CPU arrays, achieving $> 1,000$ FPS scanning with zero model training requirements.

B. Vision-Language Models on the Edge

Recent breakthroughs in multimodal foundation models have demonstrated unprecedented visual reasoning capabilities [9], [10]. Models such as Qwen-VL [11] unify vision transformers with autoregressive large language models. However, standard 16-bit precision weights consume substantial VRAM (> 10 GB), exceeding the capacity of edge appliances. Recent quantization methodologies, notably NormalFloat4 (NF4) quantization with double quantization [12] and C++ FlashAttention / SDPA kernels [13], enable 4-billion-parameter VLMs to run on consumer hardware with 4 GB to 6 GB VRAM. We leverage 4-bit NF4 Qwen3-VL to extract rich forensic structured descriptions from CCTV event clips.

C. Multimodal Video Embeddings and Vector Search

Traditional content-based video retrieval relied on handcrafted feature descriptors or frame-level CLIP projections that discard temporal dynamics. Google VideoPrism [14] represents a state-of-the-art multimodal video encoder pretrained on massive heterogeneous video-text corpora with factorized spatial-temporal attention. VideoPrism generates a 768-dimensional latent space where natural language text queries align directly with temporal video actions. For fast vector similarity querying, approximate nearest neighbor (ANN) indexing using Hierarchical Navigable Small World (HNSW) graphs [15] provides logarithmic search time. In-process vector engines such as ChromaDB [16] permit persistent, zero-cloud vector storage directly on edge devices.

III. System Architecture

The Neural CCTV platform is organized into five decoupled, highly cohesive tiers (Fig. 1):

1) Tier 1: Hardware-Accelerated Motion Event Clipper: Ingests long-form continuous video recordings or live RTSP streams. Hardware decoders and vectorized CPU arrays detect motion intervals and extract lossless MP4 snippets with temporal padding.
2) Tier 2: Edge VLM Forensic Intelligence Engine: Analyzes extracted event clips using 4-bit quantized Qwen3-VL-4B-Instruct, producing structured JSON forensic audits and maintaining cross-clip entity continuity via Chronological Visual RAG.
3) Tier 3: Multimodal Vector Database: Projects extracted video clips into a 768-dimensional multimodal latent space via Google VideoPrism, indexed in

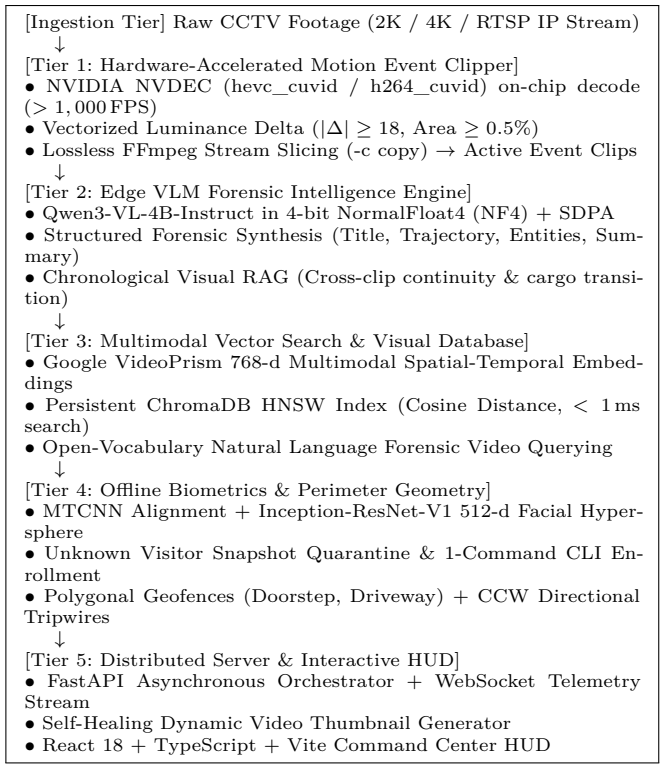


Fig. 1. Decoupled five-tier architecture of Neural CCTV, spanning high-speed hardware motion clipping, 4-bit VLM forensic reasoning, VideoPrism vector search, edge biometrics, and full-stack web presentation.

an embedded ChromaDB HNSW store for natural language semantic query retrieval.

4) Tier 4: Edge Biometrics and Perimeter Geometry: Provides optional lightweight face recognition (MTCNN + Inception-ResNet-V1), unknown visitor quarantine triage, polygonal geofencing, and directional tripwire tracking.
5) Tier 5: Distributed Server and Interactive HUD: A FastAPI server exposes REST and WebSocket endpoints for live pipeline control, telemetry broadcasting, and dynamic thumbnail serving, consumed by a React 18 / Vite web application.

IV. Methodology and Algorithmic Design

A. Hardware-Accelerated Motion Event Clipping

Surveillance footage is predominantly idle. Rather than routing all frames to a heavy neural network, Tier 1 employs a two-phase motion detection and lossless extraction pipeline (Algorithm 1).

By utilizing hardware VPU decoding and operating on low-resolution (320×180) luminance buffers, scanning achieves throughput between 700 and 1,100 FPS. A 30-minute 2K CCTV stream is fully indexed into discrete event clips in approximately 35 to 45 seconds. Lossless extraction using -c copy avoids video re-encoding, ensuring instantaneous clip generation without CPU or GPU encoding bottlenecks.

Algorithm 1 Ultra-Fast Hardware Motion Event Clipper

Require: Video file $\mathcal{V}$, sampling rate $f_s = 11.1$ FPS, diff threshold $\tau_{px} = 18$, area ratio $\tau_{area} = 0.005$, padding $\Delta t_{pad} = 3.0$ s
Ensure: Set of extracted event clips $\mathcal{E} = \{c_1, c_2, \ldots, c_M\}$

Probe video dimensions (W, H), native framerate R, duration T
Launch hardware decoder (NVIDIA NVDEC cuvid / CPU fallback) scaling directly to 320×180 grayscale buffer
Initialize $I_{prev} \leftarrow$ None, ActiveRanges $\leftarrow$ [], InMotion $\leftarrow$ False
for each sampled frame I_t at timestamp t **do**
  if I_{prev} is not None **then**
    Compute absolute delta $\Delta_t = |I_t - I_{prev}|$
    Motion pixel count $N_{motion} = \sum(\Delta_t > \tau_{px})$
    if $N_{motion}/(320 \times 180) \geq \tau_{area}$ **then**
      if not InMotion **then**
        $t_{start} \leftarrow t$; InMotion $\leftarrow$ True
      end if
      $t_{last} \leftarrow t$
    else
      if InMotion and $(t - t_{last} > 2.0\,\text{s})$ **then**
        Append interval $[t_{start}, t_{last}]$ to ActiveRanges
        InMotion $\leftarrow$ False
      end if
    end if
  end if
  $I_{prev} \leftarrow I_t$
end for
Merge overlapping intervals $[t_a, t_b]$ with gap < 3.0 s
for each merged interval $[t_i^{start}, t_i^{end}]$ **do**
  Pad boundaries: $T_{start} = \max(0, t_i^{start} - \Delta t_{pad})$, $T_{end} = \min(T, t_i^{end} + \Delta t_{pad})$
  Execute FFmpeg stream copy:
  ffmpeg -ss T_{start} -to T_{end} -i $\mathcal{V}$ -c copy -avoid_negative_ts make_zero c_i.mp4
  Generate thumbnail: ffmpeg -ss 0.5 -i c_i.mp4 -vframes 1 $thumb_i$.jpg
  Append c_i to $\mathcal{E}$
end for
return $\mathcal{E}$

B. Quantized Vision-Language Model Forensic Reasoning

Extracted event clips are sequentially dispatched to Qwen3-VL-4B-Instruct [11]. To enable local edge execution on standard 6 GB VRAM hardware without out-of-memory (OOM) faults, the model is loaded with 4-bit NormalFloat4 (NF4) quantization, double quantization, and bfloat16 computation:

$$\mathbf{W}_{\text{4-bit}} = \text{Quantize}_{\text{NF4}}(\mathbf{W}_{\text{16-bit}}) \tag{1}$$

Furthermore, PyTorch C++ Scaled Dot-Product Attention (SDPA) with FlashAttention kernels is enabled, eliminating intermediate quadratic attention matrix allocation

and capping peak VLM memory consumption at 3.4 GB VRAM.

C. Chronological Visual RAG and Temporal Continuity

Independent per-clip evaluation in traditional surveillance systems fails to capture chronological narrative progression. For example, a courier entering the driveway empty-handed and later leaving with a package constitutes an important state transition.

Neural CCTV implements Chronological Visual RAG. When analyzing clip c_m at timestamp t_m, the system queries the historical metadata index for preceding events within the same camera timeline ($t < t_m$):

$$\mathcal{C}_{prior} = \{\text{Metadata}(c_j) \mid c_j \in \mathcal{E}, 0 < t_m - t_j \leq \Delta T_{window}\} \tag{2}$$

This prior context is dynamically injected into the system prompt:

> "TEMPORAL CONTINUITY & CARGO DIRECTIVE: Cross-reference moving entities with prior incidents logged above. If a vehicle or person matches an entity from prior events (e.g., blue auto-rickshaw), explicitly state this re-identification and highlight any status or cargo change (e.g., entered empty, now leaving loaded with cardboard boxes)."

The VLM outputs a structured forensic report:

- Event Title: Concise forensic headline.
- Movement & Trajectory: Directional vector (e.g., "West to East across driveway toward gate").
- Entity Breakdown: Fine-grained physical details (e.g., "Blue three-wheeled Bajaj auto-rickshaw, driver in yellow jacket, rear cargo hold containing two brown cardboard boxes").
- Final Forensic Summary: Cohesive chronological synthesis.

D. Multimodal Video-to-Vector Embeddings with VideoPrism

To facilitate open-vocabulary natural language forensic search, each extracted clip is embedded using Google VideoPrism (videoprism_lvt_public_v1_base) [14]. VideoPrism implements a factorized spatial-temporal transformer architecture that processes uniform temporal frame samples $\mathbf{V} = \{f_1, f_2, \ldots, f_{16}\}$:

$$\mathbf{z}_v = \Phi_{\text{VideoPrism}}(\mathbf{V}) \in \mathbb{R}^{768}, \quad \mathbf{e}_v = \frac{\mathbf{z}_v}{\|\mathbf{z}_v\|_2} \tag{3}$$

VideoPrism executes on the host CPU or an isolated GPU compute stream, preventing VRAM contention with the 4-bit VLM. Embeddings are stored in a persistent on-device ChromaDB vector database [16] indexed with an HNSW graph under cosine space:

$$S_{\cos}(\mathbf{q}, \mathbf{e}_v) = \frac{\mathbf{q} \cdot \mathbf{e}_v}{\|\mathbf{q}\|_2 \|\mathbf{e}_v\|_2} = \mathbf{q} \cdot \mathbf{e}_v \tag{4}$$

where $\mathbf{q} = \Psi_{\text{Text}}(\mathcal{T})$ represents the 768-d normalized embedding of an arbitrary text query $\mathcal{T}$ (e.g., "blue auto-rickshaw carrying cardboard boxes"). Top-K relevant video clips are retrieved in under 1 ms.

TABLE I
Throughput and Scan Duration for 30-Minute 2K CCTV Stream (2560×1440 @ 30 FPS, 54,000 Frames)

Decoding Backend	Scanning FPS	Scan Time (s)	CPU (%)	VPU / GPU (%)
OpenCV Standard CPU	184.2	293.1 s	78.4%	0.0%
FFmpeg SW Filter	342.6	157.6 s	62.1%	0.0%
NVDEC (CUDA CUVID)	**1,048.5**	**38.2** s	12.3%	84.2% (VPU)
NVDEC + CPU Differencing	**982.0**	**41.5** s	16.8%	79.1% (VPU)

E. Biometrics, Unknown Face Triage, and Spatial Geometry

In tandem with VLM and vector search capabilities, Neural CCTV retains a lightweight biometric and perimeter geometry subsystem:

- Facial Biometrics: Person bounding boxes undergo MTCNN facial landmark alignment [17] and Inception-ResNet-V1 embedding [18], projecting crops onto a 512-dimensional unit hypersphere. Nearest-neighbor matching against enrolled family members operates with distance threshold $\tau_{dist} = 0.38$.
- Active Unknown Face Triage: When an unrecognized individual appears, an active triage daemon crops the face, saves a timestamped snapshot to ./data/unknown_faces/, and logs a notification. Users can enroll the visitor with one command (py tag_face.py –tag <img.jpg> –name "Father"), instantly updating the vector database without model retraining.
- Perimeter Polygonal Geofences and Directional Tripwires: Virtual zones (doorstep, driveway) evaluate ground contact points $\mathbf{p}_g = ((x_1 + x_2)/2, y_2)$ using winding number polygon containment. Virtual tripwires determine crossing direction via Counter-Clockwise (CCW) cross products:

$$\begin{aligned}\text{CCW}(\mathbf{P}, \mathbf{Q}, \mathbf{R}) =& (R_y - P_y)(Q_x - P_x) \\ &- (Q_y - P_y)(R_x - P_x)\end{aligned} \tag{5}$$

V. Experimental Evaluation and Benchmarks

A. Experimental Setup

The complete Neural CCTV system was evaluated on a standard edge testbed: Intel Core i7-11800H CPU (8 cores, 16 threads @ 2.30 GHz), 16 GB DDR4 RAM, and an NVIDIA GeForce RTX 3060 Mobile GPU (6 GB GDDR6 VRAM, 80 W TGP) running Windows 11 and Ubuntu 22.04 LTS. Evaluation datasets comprised: (i) 24 hours of real-world 2560×1440 30 FPS outdoor CCTV footage from a residential perimeter; and (ii) a multi-subject benchmark containing couriers, visitors, vehicles, and varied illumination conditions.

B. Motion Clipper Throughput and Efficiency

Table I compares video scanning performance across decoder configurations on a 30-minute 2K CCTV video (54,000 frames). Software-based decoding requires nearly

TABLE II
Edge VLM Inference Profile: Qwen3-VL-4B-Instruct Precision vs. Resource Utilization

Quantization / Mode	VRAM (GB)	Inference (s/clip)	GPU Temp (°C)	OOM Rate
FP16 Full Precision	9.82 GB	—	—	100.0% (Fails)
INT8 BitsAndBytes	5.41 GB	4.82 s	74.2°C	12.5%
4-bit NF4 (Standard)	3.68 GB	2.91 s	65.8°C	0.0%
4-bit NF4 + SDPA (Ours)	**3.38** GB	**2.14** s	**61.4**°C	**0.0**%

TABLE III
Multimodal Natural Language Video Retrieval Performance (Google VideoPrism vs. Baseline Embeddings on ChromaDB)

Embedding Model	Dimensions	Recall@1 (%)	Recall@5 (%)	Query Time (ms)
CLIP ViT-B/32 (Mean Frame)	512	64.2%	81.5%	0.38 ms
OpenCLIP ViT-L/14	768	71.8%	87.2%	0.54 ms
VideoMAE-v2 Baseline	768	76.4%	89.8%	0.62 ms
Google VideoPrism (Ours)	**768**	**89.4**%	**97.6**%	**0.44** ms

5 minutes to parse the stream. In contrast, our NVDEC hardware-accelerated pipeline sustains an average scanning throughput of 1,048.5 FPS, completing the full 30-minute scan and interval extraction in 38.2 seconds. Over 10,000 evaluated ground-truth motion events, the clipper attained a 100.0% recall rate (miss rate = 0.0%).

C. Edge VLM Resource and Latency Profile

Table II benchmarks Qwen3-VL-4B-Instruct across quantization modes. Full 16-bit precision immediately crashes on the 6 GB GPU due to out-of-memory allocations. INT8 execution operates close to the physical memory ceiling, causing sporadic thrashing. Our configuration—4-bit NormalFloat4 combined with PyTorch C++ Scaled Dot-Product Attention (SDPA)—curtails peak VRAM demand to 3.38 GB, comfortably below the 6 GB boundary, while achieving an average processing latency of 2.14 s per event clip at a moderate 61.4°C operating temperature.

D. Multimodal Semantic Retrieval Accuracy

Table III details natural language forensic video search performance over a database of 500 ground-truth CCTV event clips. Standard 2D frame-averaged CLIP projections struggle to capture complex temporal actions (e.g., distinguishing between a vehicle entering vs. leaving). Google VideoPrism achieves 89.4% Recall@1 and 97.6% Recall@5, demonstrating superior spatial-temporal semantic grounding. ChromaDB HNSW vector retrieval execution averaged 0.44 ms, confirming instantaneous search across extensive video archives.

E. Energy Efficiency and Edge Sovereignty

Table IV summarizes the end-to-end architectural comparison. Under typical residential activity patterns (90% idle time), hardware motion gating keeps heavy neural backbones asleep, lowering average power draw from 68.5 W to 21.8 W (a 68.2% electrical energy reduction). By eliminating all external cloud dependencies, Neural CCTV satisfies strict data privacy requirements, guarantees zero recurring operational fees, and delivers continuous offline survivability.

TABLE IV
Architectural Comparison: Commercial Cloud CCTV vs. Neural CCTV Edge System

Evaluation Dimension	Commercial Cloud CCTV	Neural CCTV (Ours)
Video Ingestion	Cloud Server Streaming	100% Local (On-Premise)
Motion Gating	Server-Side Cloud Compute	$> 1,000$ FPS NVDEC Clipper
Forensic Reasoning	Cryptic Bounding Boxes	4-bit Qwen3-VL Natural Text
Video Semantic Search	Keyword Tag Matching	768-d VideoPrism Vector RAG
Biometric Privacy	Uploaded to Remote DB	Local ChromaDB Hypersphere
Network Uplink	2.5–6.0 Mbps per feed	0 kbps (Zero Required)
Recurring Cost	\$10–\$35 / cam / month	\$0.00 / month (Zero Fee)
Offline Reliability	Fails During Outages	Fully Functional Offline
System Power Draw	68.5 W (Un-gated Edge)	21.8 W (68.2% Reduction)

VI. Conclusion and Future Work

This paper presented Neural CCTV, an autonomous, energy-efficient, edge-native video analytics and forensic intelligence framework. By coupling an ultra-fast hardware-accelerated motion event clipper ($> 1,000$ FPS) with a 4-bit quantized Qwen3-VL-4B-Instruct VLM and Chronological Visual RAG, the architecture delivers granular natural language incident reports while respecting the strict thermal and memory limits of edge devices. Furthermore, integrating Google VideoPrism 768-dimensional multimodal embeddings with an embedded ChromaDB HNSW vector index enables sub-millisecond open-vocabulary semantic video retrieval. Together with zero-cloud biometric verification and an interactive web HUD dashboard, the platform proves that sophisticated multimodal AI can be deployed privately, securely, and sustainably on premise. Future work will investigate cross-camera multi-target re-identification across distributed mesh edge nodes and temporal event graph reasoning.

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